General Dessert #8: Digital Operating Modes and Procedures

RTTY/FSK, FT8 and the JT/FT family, PACTOR/VARA, AREDN mesh, Winlink, sideband conventions, and band-plan locations for digital modes.

Group G2E · 14 pool questions · Self-study reference · Practice exams at eham.net/exams

Key Facts Reference

Sideband and Tone Conventions

Topic	Answer	Q
RTTY via AFSK with an SSB transmitter	**LSB.** Historic convention; reversed tones decode as garbage.	G2E01
Standard sideband for JT65/ JT9/FT4/FT8 via AFSK	**USB.** All WSJT-X modes assume USB.	G2E05
Most common RTTY shift on HF	**170 Hz** between mark and space.	G2E06
Can't decode RTTY/FSK that appears tuned in	**All of the above**: mark/space reversed, wrong baud rate, or wrong sideband.	G2E14

FT8 and the JT/FT Family

Topic	Answer	Q
Answering a CQ on FT8: where to transmit?	**Find a clear frequency in the alternate time slot** (15-second slot opposite the caller). Lets others copy both ends.	G2E04
FT8 requirement	**Computer time accurate to within about 1 second** – modes are slot-synchronized.	G2E07
Common FT8 location on 20 m	**Approximately 14.074 MHz to 14.077 MHz.**	G2E15
Digital segment on 20 m generally	**14.070 MHz to 14.100 MHz.**	G2E08

PACTOR, VARA, Winlink, and AREDN

Topic	Answer	Q
Symptoms of QRM on PACTOR/VARA	**All of the above**: retries/timeouts, long pauses, failure to connect. ARQ modes require clean links – QRM triggers retries, not graceful recovery.	G2E03
Joining a two-station PACTOR contact	**Not possible** – PACTOR is a point-to-point ARQ link, only two stations.	G2E09
Contacting a digital messaging gateway	**Transmit a connect message on the station's published frequency.**	G2E10
AREDN mesh purpose	**High-speed data services during emergency or community events.** IP-style mesh on amateur microwave.	G2E11
What is Winlink?	**All of the above**: amateur wireless email-to-internet network, descendant of Packet, both VHF and HF.	G2E12
Other name for a Winlink Remote Message Server	**Gateway.** The RF-to-internet bridge node.	G2E13

Dessert 8 Question Bank (G2E)

G2E01. Which mode is normally used when sending RTTY signals via AFSK with an SSB transmitter?
A. USB
B. DSB
C. CW
D. LSB

G2E03. What symptoms may result from other signals interfering with a PACTOR or VARA transmission?
A. Frequent retries or timeouts
B. Long pauses in message transmission
C. Failure to establish a connection between stations
D. All these choices are correct

G2E04. Which of the following is good practice when choosing a transmitting frequency to answer a station calling CQ using FT8?
A. Always call on the station's frequency
B. Call on any frequency in the waterfall except the station's frequency
C. Find a clear frequency during the same time slot as the calling station
D. Find a clear frequency during the alternate time slot to the calling station

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G2E05. What is the standard sideband for JT65, JT9, FT4, or FT8 digital signal when using AFSK?
A. LSB
B. USB
C. DSB
D. SSB

G2E06. What is the most common frequency shift for RTTY emissions in the amateur HF bands?
A. 85 Hz
B. 170 Hz
C. 425 Hz
D. 850 Hz

G2E07. Which of the following is required when using FT8?
A. A special hardware modem
B. Computer time accurate to within approximately 1 second
C. Receiver attenuator set to −12 dB
D. A vertically polarized antenna

G2E08. In what segment of the 20-meter band are most digital mode operations commonly found?
A. At the bottom of the slow-scan TV segment, near 14.230 MHz
B. At the top of the SSB phone segment, near 14.325 MHz
C. In the middle of the CW segment, near 14.100 MHz
D. Between 14.070 MHz and 14.100 MHz

G2E09. How do you join a contact between two stations using the PACTOR protocol?
A. Send broadcast packets containing your call sign while in MONITOR mode
B. Transmit a steady carrier until the PACTOR protocol times out and disconnects
C. Joining an existing contact is not possible, PACTOR connections are limited to two stations
D. Send a NAK code

G2E10. Which of the following is a way to establish contact with a digital messaging system gateway station?
A. Send an email to the system control operator
B. Send QRL in Morse code
C. Respond when the station broadcasts its SSID
D. Transmit a connect message on the station's published frequency

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G2E11. What is the primary purpose of an Amateur Radio Emergency Data Network (AREDN) mesh network?
A. To provide FM repeater coverage in remote areas
B. To provide real time propagation data by monitoring amateur radio transmissions worldwide
C. To provide high-speed data services during an emergency or community event
D. To provide DX spotting reports to aid contesters and DXers

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G2E12. Which of the following describes Winlink?
A. An amateur radio wireless network to send and receive email on the internet
B. A form of Packet Radio
C. A wireless network capable of both VHF and HF band operation
D. All these choices are correct

G2E13. What is another name for a Winlink Remote Message Server?
A. Terminal Node Controller
B. Gateway
C. RJ-45
D. Printer/Server

G2E14. What could be wrong if you cannot decode an RTTY or other FSK signal even though it is apparently tuned in properly?
A. The mark and space frequencies may be reversed
B. You may have selected the wrong baud rate
C. You may be listening on the wrong sideband
D. All these choices are correct

G2E15. Which of the following is a common location for FT8?
A. Anywhere in the voice portion of the band
B. Anywhere in the CW portion of the band
C. Approximately 14.074 MHz to 14.077 MHz
D. Approximately 14.110 MHz to 14.113 MHz

Answer Key

G2E01: D	G2E03: D	G2E04: D	G2E05: B
G2E06: B	G2E07: B	G2E08: D	G2E09: C
G2E10: D	G2E11: C	G2E12: D	G2E13: B
G2E14: D	G2E15: C		